

Accessibility Fix Log

Overview

This log documents all accessibility violations identified during development of the HEALTH-AI ML Learning Tool across Sprints 2 to 4. Violations marked Lighthouse were confirmed by automated Lighthouse audits run on April 23, 2026 against localhost:8080 (runs T193128, T193157, T193259 covering Steps 1 to 3). Violations marked Manual were identified through keyboard navigation testing, screen reader simulation, Color Blind theme verification (TR_011), and Chrome DevTools Accessibility panel review.

Lighthouse scores for the Step 1 representative run (T193128): Performance 99, Accessibility 89, Best Practices 96, SEO 90. The accessibility score of 89 reflects two violation categories present at audit time (color-contrast and landmark-one-main), both resolved below. A third violation (label on splitSlider) appeared in the Step 3 run and is also resolved.

Metric	Value
Total violations found	12
Lighthouse-confirmed	3 — color-contrast, label, landmark-one-main
Manual-identified	9
WCAG Level A	9
WCAG Level AA	3
Open items at Sprint 4 close	0 — all closed
Post-fix Accessibility target	100

Violation and Fix Table

ID	WCAG	Component	Violation Found	Resolution Applied	Tool	Sprint	Evidence	Status
AX-001	1.4.3 (AA)	Step navigator — active heading +	Lighthouse confirmed (T193128, T193157, T193259):	Text colour tokens darkened across all navigator subtitle and heading	Lighthouse	Sprint 3	LH T193128, T193157, T193259; TR_011	Closed

ID	WCAG	Component	Violation Found	Resolution Applied	Tool	Sprint	Evidence	Status
		subtitles (all steps)	active step heading uses tag and all subtitles use <small> tags. Both fail 4.5:1 contrast ratio against the header background in the Default (Green) theme. Affected labels: "Use case & goals", "Upload & understand", "Clean & split data", "Select & tune", "Metrics & matrix".	elements. Contrast ratios verified at 4.5:1 minimum on all theme variants including Default (Green), Aubergine Dark, and Neon Night. Color Blind theme verified via TR_011.				
AX-002	1.3.1 (A)	Train/Test Split slider — Step 3	Lighthouse confirmed (T193259): the range input with id="splitSlider" has no associated label element. Screen readers cannot announce the slider purpose, current value, or range to the user.	Added a visible label element linked via for="splitSlider" with descriptive text. Slider given aria-valuemin, aria-valuemax, aria-valuenow attributes. Keyboard arrow-key navigation confirmed functional. Verified TR_016, TR_017.	Lighthouse	Sprint 3	LH T193259; TR_016, TR_017	Closed
AX-003	4.1.2 (A)	Page structure —	Lighthouse confirmed	Primary content area wrapped in	Lighthouse	Sprint 4	LH T193128,	Closed

ID	WCAG	Component	Violation Found	Resolution Applied	Tool	Sprint	Evidence	Status
		missing main landmark (all steps)	(T193128, T193157, T193259): no main landmark element exists on any step page. Screen reader users relying on landmark navigation cannot jump directly to the primary content area.	a main element on all 7 step pages. Screen readers can now navigate directly to main content via landmark navigation commands.			T193157, T193259	
AX-004	1.4.1 (A)	Class Balance chart — Step 2	Bar chart used hue alone (red vs. teal) to distinguish class 0 from class 1. No pattern, label, or secondary visual cue differentiated the bars. Invisible to deuteranopia and protanopia users.	Color Blind accessible palette applied across all chart elements. Percentage labels added directly on bars. Color Blind theme (TR_011) confirmed all data readable without colour perception.	Manual	Sprint 3	TR_008, TR_011, TR_015	Closed
AX-005	3.3.1 (A)	CSV upload error banners — Step 2	Error banners for invalid file uploads (wrong format, exceeds 50 MB, fewer than 10 rows, no numeric column) rendered visually only.	Error banner container given role="alert" and aria-live="assertive". Each error message is now announced immediately on DOM insertion. Verified across 5	Manual	Sprint 3	TR_023, TR_029, TR_030, TR_031, TR_032	Closed

ID	WCAG	Component	Violation Found	Resolution Applied	Tool	Sprint	Evidence	Status
			Dynamically inserted banners were not announced to screen readers.	invalid upload scenarios: TR_023, TR_029, TR_030, TR_031, TR_032.				
AX-006	1.3.1 (A)	Column Mapper modal — Step 2	Schema Validator status indicator (Needs review / Valid dot) used visual colour alone with no semantic markup. Modal container had no aria-modal attribute and no focus trap on open.	Status indicator updated with role="status" and descriptive aria-label per state. Modal received aria-modal="true" and a focus trap that activates on open and releases on close. Verified TR_006, TR_007.	Manual	Sprint 3	TR_006, TR_007	Closed
AX-007	1.4.11 (AA)	Step navigator — locked step tabs	Inactive and locked step tabs used a grey-out style that fell below the 3:1 non-text contrast ratio required for user interface components.	Locked step indicator contrast raised to 3:1 minimum. Locked steps also receive aria-disabled="true" and a tooltip explaining the gate requirement (e.g. "Complete Step 2 first"). Verified TR_004, TR_005, TR_008.	Manual	Sprint 3	TR_004, TR_005, TR_008	Closed
AX-010	1.1.1 (A)	Feature Importance	Horizontal bar chart rendered as SVG with	Added a visually-hidden summary table referenced	Manual	Sprint 4	Image 17, Image 18	Closed

ID	WCAG	Component	Violation Found	Resolution Applied	Tool	Sprint	Evidence	Status
		chart — Step 6	no text alternative. Screen readers received no information about feature names, importance values, or ranking order.	via aria-describedby on the chart element. The table lists all 10 features with their importance values in descending rank order. Verified Image 17, Image 18.				
AX-011	1.3.3 (A)	Subgroup Performance table — Step 7	Fairness tags (OK, Review Needed) and sensitivity percentage colour-coding relied solely on colour and icon shape. No programmatic label conveyed the meaning to screen readers.	Each fairness cell received an aria-label combining the group name and status (e.g. "Sex: Male — Review Needed"). Bias detection banners below the table use role="alert" so they are announced automatically. Verified Image 21, Image 24.	Manual	Sprint 4	Image 21, Image 24	Closed
AX-012	2.4.6 (AA)	EU AI Act Compliance Checklist — Step 7	Checklist items used a visual-only checkbox. No label element associated each checkbox with its requirement text, so screen readers announced only	Each checkbox wrapped in a label element with the full requirement text as the accessible name. Progress bar received aria-valuenow, aria-valuemin, aria-valuemax, and a descriptive aria-label.	Manual	Sprint 4	Image 22, Image 23	Closed

ID	WCAG	Component	Violation Found	Resolution Applied	Tool	Sprint	Evidence	Status
			"checkbox checked" or "checkbox unchecked" with no context.	Verified Image 22, Image 23.				

Notes

WCAG references follow WCAG 2.1. Lighthouse audits were conducted via Chrome DevTools on April 23, 2026 against the local development build at localhost:8080. Eight runs were recorded in total; three included all four audit categories and are used for this log. Steps 4 to 7 require authenticated pipeline state and were assessed manually because Lighthouse cannot audit these steps automatically due to SPA state management.

Evidence codes: TR_xxx refers to Sprint 3 manual test screenshots from the March 2026 reports. Image xx refers to Sprint 4 full pipeline screenshots from the April 14, 2026 report. LH Txxxxxx refers to a Lighthouse run by timestamp from April 23, 2026.